

Algorithmic Description of Free-Fermionic Line-Graph States

Jannis Ruh

5th August 2026

We give details on the efficient description and parametrisation of the eigenstates of Hamiltonians whose commutation frustration graph is a line graph. This problem has been largely solved by Ref. [1], however, to actually implement such states in code, it is beneficial to have a more detailed algorithmic description that fills the gaps.

Contents

1. Definitions and Notation	2
1.1. Basics	2
1.1.1. Linear Algebra	2
1.1.2. Group Theory	2
1.1.3. Graph Theory	2
1.2. Basis Operators	3
2. Problem Setup	4
3. Algorithmic Description	5
3.1. Majorana Mapping	5
3.2. Diagonalisation of the Majorana Hamiltonian	10
3.2.1. Diagonalisation and Preparation of Eigenstates of the Problem Hamiltonian	12
<i>APPENDICES</i>	15
A. Minor Things	15
B. Symmmetries Under the Ψ and ψ Mappings	16
B.1. Why ψ is not Injective in General	16
B.2. Elements in $\psi^{-1}(\mathbf{1})$ must be an Effective T-join (or a Cycle)	16
B.3. How to Make ψ Injective	16
C. On the Schur Decomposition	17
D. On the Generalised Euler Angles Decomposition	18
Bibliography	19

1. Definitions and Notation

1.1. Basics

1.1.1. Linear Algebra

Let $\mathcal{H}$ be a Hilbert space. We denote the inner product by $\langle \cdot, \cdot \rangle$, and define it to be sesquilinear in the first argument. The canonical norm, $\|\cdot\|$, is the one induced by the inner product. Let $L(\mathcal{H})$, $\mathcal{L}(\mathcal{H})$ and $U(\mathcal{H})$ be the space of linear mappings, bounded linear mappings and unitary subgroup on $\mathcal{H}$, respectively. Given $T \in \mathcal{L}(\mathcal{H})$, we denote its Hilbert-adjoint by $T^\dagger$. If $\mathcal{H}$ is finite dimensional, i.e., $\mathcal{H} \cong \mathbb{C}^d$ for some $d \in \mathbb{N}$, we often use $\mathcal{H}$ and $\mathbb{C}^d$ synonymously, and similarly for $L(\mathcal{H}) \cong M_d = M_d(\mathbb{C})$, where the letter is the space of complex $d \times d$ matrices, and subspaces or subgroups thereof, i.e., for $\mathcal{L}(\mathcal{H}) \cong GL_d = GL_d(\mathbb{C})$ and $U(\mathcal{H}) \cong U(d)$. In that case, we also define the Hilbert-Schmidt inner product as

$$\langle \cdot, \cdot \rangle_{\text{HS}} : \mathcal{L}(\mathcal{H}) \times \mathcal{L}(\mathcal{H}) \rightarrow \mathbb{C}, (A, B) \mapsto \text{Tr}(A^\dagger B), \quad (1.1.1)$$

where $\text{Tr}(\cdot)$ denotes the trace, and the corresponding induced norm is denoted as $\|\cdot\|_{\text{HS}}$.

1.1.2. Group Theory

Let $(G, \cdot)$ be a group that acts on a set M from left and right. We write the (group) commutator as

$$[\cdot, \cdot] : G \times G \rightarrow G, (g, h) \mapsto ghg^{-1}h^{-1}, \quad (1.1.2)$$

and the conjugation action, $g * x$, is defined as

$$* : G \times M \rightarrow M, (g, x) \mapsto gxg^{-1}. \quad (1.1.3)$$

For a subset $X \subseteq G$, $\langle g \in X \rangle$ denotes the subgroup generated by the elements in X . The notation $H \leq (<) G$ implies that H is a subgroup of G .

1.1.3. Graph Theory

Let V be a finite set and $E \subseteq (V \times V) \setminus \{(v, v) \mid v \in V\}$. Then the tuple $G = (V, E)$ defines a simple graph with vertex set V and edge set E . If the graph is directed, we have the convention that an edge $(u, v) \in E$ goes from u to v . If not stated otherwise, all graphs are undirected. For $X \subseteq V$, we denote $G[X] = (X, E \cap (X \times X))$ as the induced subgraph by X . The open neighbourhood of a vertex $v \in V$ is defined is $\mathcal{N}(v) = \{u \in V \mid (u, v) \in E\}$. We are specifically interested in line graphs:

Definition 1.1.1 (Line graph) *A graph $G = (V, E)$ is a line graph if there exists a graph $R = (V', E')$ such that $V = E'$ and $E = \{(e_1, e_2) \in E' \times E' \mid |e_1 \cap e_2| = 1\}$ (where the intersection of the edges is taken as if they were sets). The graph R is called the root graph of G and vice versa we write $G = \mathcal{L}(R)$ as the line graph of R .*

Analogously, line graphs are characterised by that they contain no induced subgraph that is one of the nine forbidden graphs shown in Fig. 1.1.1.

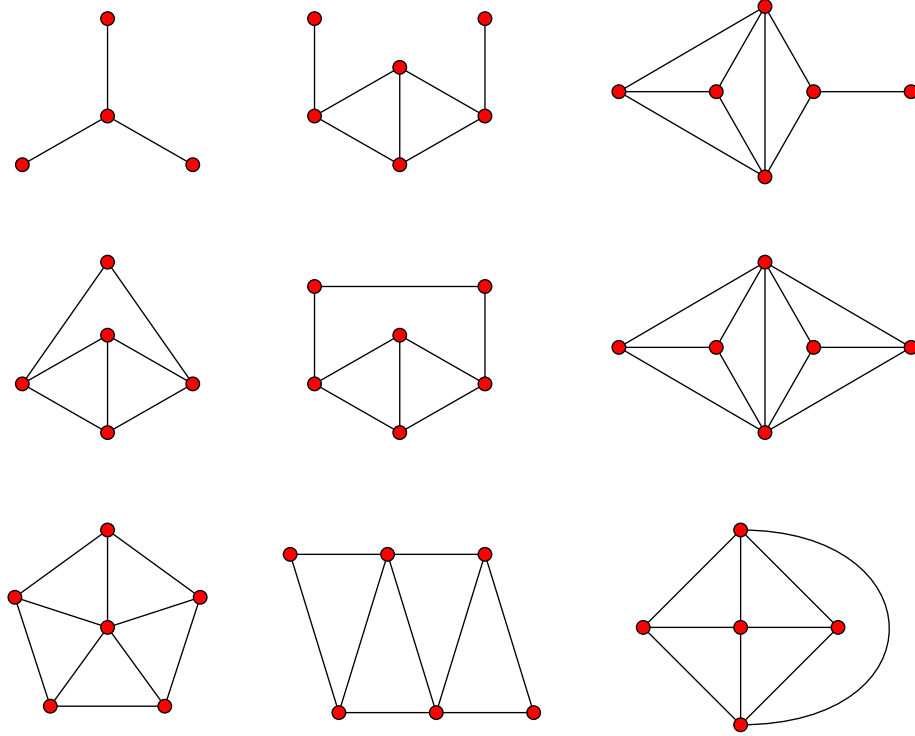


Figure 1.1.1.: The nine minimal non-line graphs [2].

Definition 1.1.2 (Spanning tree) Let $G = (V, E)$ be a connected graph. A graph $T = (V', F)$ is a spanning tree of G if it is a tree.

1.2. Basis Operators

Definition 1.2.1 (Pauli group) Let $n \in \mathbb{N}$ and $Z_i, X_i \in \mathbb{C}^{2^n \times 2^n}$, $i \in \{1, \dots, n\}$, denote the standard Z and X Pauli matrices on n spins. We define the Pauli group in its standard representation as

$$\mathcal{P}_n = \langle i, Z_j, X_j \mid j \in \{1, \dots, n\} \rangle. \quad (1.2.1)$$

Definition 1.2.2 (Majorana group) Let $n \in \mathbb{N}$ and $\gamma_i \in \mathbb{C}^{2^n \times 2^n}$, $i \in \{1, \dots, 2n\}$, denote Majorana matrices (e.g., via the standard Jordan-Wigner mapping) for n fermions. We define the Majorana group in that representation as

$$\mathcal{W}_n = \langle i, \gamma_j \mid j \in \{1, \dots, 2n\} \rangle. \quad (1.2.2)$$

Definition 1.2.3 (Abstract Majorana group) *Let $m \in \mathbb{N}$ and $\gamma_i \in \mathcal{H}$ be hermitian and unitary with $\llbracket \gamma_i, \gamma_j \rrbracket = -1$ for $i, j \in \{1, \dots, m\}$ with $i \neq j$. We define the abstract - and degenerate¹ iff n is odd - Majorana group as*

$$\mathcal{W}^m = \langle i, \gamma_j \mid j \in \{1, \dots, m\} \rangle. \quad (1.2.3)$$

Definition and Proposition 1.2.4 (Generalised binary group) *We define $\mathcal{R} \subset U(\mathcal{H})$ to be a group where for all $g, h \in \mathcal{R}$ it holds that*

$$\llbracket g, h \rrbracket \in \{-1, +1\}. \quad (1.2.4)$$

Furthermore, we require that all elements, except the identity, have a vanishing trace.

Let $X = \langle g_1, \dots, g_n \rangle \leq \mathcal{R}$, $n \in \mathbb{N}$, such that $g^2 \in A$ for all $g \in X$, where $A < X$ is a subgroup of the complex units, i.e., $A \leq \mathbb{C} \setminus \{0\}$. Define $W \in \mathbb{Z}_2^n$ via $W_{ij} = (\log_{-1}(\llbracket g_i, g_j \rrbracket)) \in \{0, 1\}$ for $i, j \in \{1, \dots, n\}$ and the remaining entries to be zero. Let F be a fixed set of representatives of elements in $A/\{-1, +1\}$ with $1 \in F$. We define the group $\mathcal{H} = (F, Z_2, Z_2^n, \cdot)$ via

$$\begin{aligned} \cdot : \mathcal{H} \times \mathcal{H} &\rightarrow \mathcal{H}, \\ (a, p, x), (b, q, y) &\mapsto (r(ab), u(ab) + p + q + x^\top W y, x + y), \end{aligned} \quad (1.2.5)$$

where $r : A \rightarrow F$, $u : A \rightarrow \mathbb{Z}_2$ such that $a = r(a)(-1)^{u(a)}$ for all $a \in A$. Given $(a, p, x) \in \mathcal{H}$, its inverse is $(r(a^{-1}), u(a^{-1}) + p + x^\top W x, x)$. The mapping

$$\mu : \mathcal{H} \rightarrow X, (a, p, x) \mapsto a(-1)^p g_1^{x_1} \cdots g_n^{x_n} \quad (1.2.6)$$

is an isomorphism. Setting $\Omega = W + W^\top$, we have for all $(a, p, x), (b, q, y) \in \mathcal{H}$

$$\llbracket \mu(a, p, x), \mu(b, q, y) \rrbracket = (-1)^{x^\top \Omega y}. \quad (1.2.7)$$

Proof. It is an easy exercise to show that the mappings r and u are well-defined and unique (as F is fixed) and $\mathcal{H}$ is indeed a group with the according inverse elements (use the uniqueness of the $a = r(a)(-1)^{u(a)}$ decomposition for $a \in A$). An elementary calculation then shows that μ is a homomorphism. Per construction, μ is then also clearly bijective. Another elementary calculation shows the commutator property. $\square$

2. Problem Setup

Let $\mathcal{H}$ be finite-dimensional complex Hilbert space, i.e., $\mathcal{H} \cong \mathbb{C}^d$ for some $d \in \mathbb{N}$.

Definition 2.0.1 (Hamiltonian) *Let $V \subseteq \mathcal{R}$ be an hermitian subset, we study generic Hamiltonians of the form*

$$H = \sum_{g \in V} w_g g, \quad (2.0.1)$$

¹In the sense that the commutator matrix $(\log_{-1}(\llbracket \gamma_i, \gamma_j \rrbracket))_{1 \leq i, j \leq m} \in \{0, 1\}$ (cf. Def. 1.2.4) is not non-degenerate over $\mathbb{Z}_2$, i.e., it is not a symplectic form anymore.

with $w_g \in \mathbb{R} \setminus \{0\}$ for all $g \in V$. Without loss of generality, we assume that for all $g \in V$ there is no $h \in V$ with $g \propto h$ as well as $-1 \notin V$.

Definition 2.0.2 (Frustration graph) *Let H be as in Def. 2.0.1. Its frustration graph is defined as $G := \mathcal{F}(H) := \mathcal{F}(V) := (V, E)$, where the edge set is given by*

$$E = \{(g, h) \in V \times V \mid \llbracket g, h \rrbracket = -1\}, \quad (2.0.2)$$

that is, vertices are neighbouring if, and only if, the corresponding operators anticommute.

Problem 2.0.3 (Problem Statement) *Let V and H be as in Def. 2.0.1 such that the associated frustration graph G is a line graph. We want to find an efficient diagonalisation parametrisation and procedure of the Hamiltonian.*

Remark 2.0.4 *To solve Prob. 2.0.3 one may have to specialise the group $\mathcal{R}$, e.g., to $\mathcal{P}$ (or maybe $\mathcal{W}$), in the final step. Furthermore, while we can find a diagonalisation procedure, it may be not trivial to find the ground state energy (as we don't have a simple parametrisation of the eigenvalues).*

Remark 2.0.5 *We have $V \subset X = \langle i, g \in V \rangle$ where X fulfills the conditions as in Def. 1.2.4. This gives us a numerically convenient encoding via $\mathcal{H}$. Especially the third component of $\mathcal{H}$, Z_2^n , for some $n \in \mathbb{N}$, is vectorspace over $\mathbb{Z}_2$ and independence in X is equivalent to linear independence $\mathbb{Z}_2^n$ (up to a potential phase).*

3. Algorithmic Description

Let H and $G = (V, E)$ be as in Prob. 2.0.3. To solve Prob. 2.0.3, we follow [1], filling in the gaps and giving mathematical details as well as some implementation options. Note that we do not require G to be connected (although this would make things easier in an implementation).

3.1. Majorana Mapping

1. Find the root graph $R = (V', E')$ of G , i.e., $G = \mathcal{L}(R)$ and $E' = f(V)$ for some bijective $f : V \rightarrow E'$. *The Sagemath library has an implementation for that: 'sage.graphs.line_graph.root_graph(...)'; the complexity is polynomial (they basically calculate the Krausz decomposition which boils down to finding maximal cliques which is clearly polynomial for line graphs). The function returns the root graph R and the bijective mapping f from the V to E' .*
2. We allow $m := |V'|$ to be odd and R to be disconnected, where we write $R_1 = (V'_1, E'_1), \dots, R_N = (V'_N, E'_N)$, $N \in \mathbb{N}$, for the connected components of R .
3. Label the edges in E' arbitrarily with $\{1, \dots, |E'|\}$.
4. To each vertex $v \in V'$ we assign a (abstract) Majorana mode $\gamma_v \in \mathcal{W}^m$. Furthermore define the vector $\gamma = (\gamma_1, \dots, \gamma_m)^\top$.

5. Define the injective map

$$\phi : V = f^{-1}(E') \rightarrow \mathcal{W}^m, \quad g = f^{-1}((u, v)) \mapsto (-1)^{b((u, v))} i \gamma_u \gamma_v, \quad (3.1.1)$$

where we specify $b((u, v)) \in \{0, 1\}$ later, such that we can define an inverse of ϕ that is also a homomorphism. For convenience, we may write $f^{-1}(\cdot, \cdot) = f^{-1}((\cdot, \cdot))$ and $b(\cdot, \cdot) = b((\cdot, \cdot))$.

6. Note that, per construction, ϕ preserves the commutator (not the multiplication though), i.e., $[[g, h]] = [[\phi(g), \phi(h)]]$.
7. Construct a spanning tree $T = (V', F)$ of R where $F \subseteq E'$, or rather construct a spanning tree $T_i = (V'_i, F_i)$ for each component R_i for $i \in \{1, \dots, N\}$ and define T to be the union of those trees, i.e., $T = (V' = \bigcup_{i=1}^N V'_i, F = \bigcup_{i=1}^N F_i)$. *If the vertices V have weights associated to them, it may be convenient to construct a maximum or minimum spanning tree; this can be done using, for example, the Dijkstra-Jarnik-Prim algorithm. Sagemath might also have this implemented. Otherwise, here is a very basic algorithm: Start with $C_c = C_n = F = \emptyset$ and set $U = V'$. Pick a root $r \in V'$ and update $U = U \setminus \{r\}$ and $C_c = \{r\}$. Now, while $U \neq \emptyset$ (U -loop) do: while $C_c \neq \emptyset$ (C_c -loop) do: pick a vertex $v \in C_c$ and update $C_c = C_c \setminus \{v\}$ and for each $u \in \mathcal{N}(v) \cap U$, update $U = U \setminus \{u\}$ and $C_n = C_n \cup \{u\}$ and $F = F \cup \{(u, v)\}$. After C_c -loop, update $C_c = C_n$ and $C_n = \emptyset$. After U -loop, $T = (V', F)$ is a spanning tree of R . This runs in linear time in $|V'|$.*
8. For each edge $(u, v) \in F$ choose $b(u, v)$ arbitrarily, e.g., we set $d(u, v) = 0$.
9. Cycles in R and G introduce a problem we have to solve to be able to define an inverse of ϕ as a homomorphism, as they introduce multiple paths to connect two vertices. We need to “fix the action” of a cycle so that they cannot violate the homomorphism. This can be done via the fundamental cycle symmetries which are shown to generate all cycles [1]:
10. Define the set S' of fundamental cycle symmetry generators as follows. Set $S' = \{\}$ and do the following paragraph w.r.t. each component $i \in \{1, \dots, N\}$:
 Fix a root $r \in V'_i$ in the spanning tree T_i . For each $e' = (u, v) \in E'_i \setminus F_i$ let $P(u, r) \subset E'_i$ and $P(v, r) \subset E'_i$ be the unique paths in T_i from u and v to r , respectively, and define the path $P(e' = (u, v)) = P(u, r) \Delta P(v, r) \Delta \{(u, v)\}$ where Δ denotes the symmetric difference.
 Now, in the whole graph, for all $e' \in E' \setminus F$, define

$$\eta(e') = i^{|P(e')| \bmod 2} \prod_{e \in P(e')} f^{-1}(e) \quad (3.1.2)$$

where we fix the order in the product to have increasing labels from left to right. The additional phase ensures that the product is hermitian¹ (cf. Prop. A.0.1). Append $\{\eta(e') \mid e' \in E' \setminus F\}$ to S' . Note that η is bijective from $E' \setminus F$ to S' .

¹In [1] this phase is more specific - it enforces the operator string to be of the form $p_1 \cdot p_l$ with $p_1, \dots, p_l \in \{X, Y, Z\}$, $l \in \mathbb{N}$ - however, this changes only a sign which does not matter since we consider all possible signs later on anyways.

11. The elements of S' are symmetries, i.e., they commute with each other and every term in the Hamiltonian. Indeed, using Prop. A.0.2, we have for $g \in V$ and $e' \in E' \setminus F$

$$\llbracket g, \eta(e') \rrbracket = \prod_{e \in P(e')} \llbracket g, f^{-1}(e) \rrbracket \quad (3.1.3)$$

$$= \prod_{e \in P(e')} \llbracket \phi(g), \phi(f^{-1}(e)) \rrbracket \quad (3.1.4)$$

$$= \left[\phi(g), \prod_{e \in P(e')} \phi(f^{-1}(e)) \right] \quad (3.1.5)$$

$$= \llbracket \phi(g), \mathbb{1} \rrbracket \quad (3.1.6)$$

$$= \mathbb{1} , \quad (3.1.7)$$

as a cycle in the Majorana picture is clearly proportional to the identity (each vertex in the line graph appears exactly twice).

12. Define $S = \{s \in S \mid s \not\propto \mathbb{1}\}$ and $S_I = S' \setminus S$. Clearly, S is a set of stabilisers.
13. Find an independent generating set of S (cf. Rem. 2.0.5). In Ref. [1] a single component is considered and it is stated that S' is already independent up to signs. However, I am not sure whether this easily generalises to multiple components and furthermore there are indeed examples where S_I is non-empty. Therefore, we just generically find an independent generating set of S . Let $S_0 \subseteq S$ be an independent generating set, and set $S_1 = S \setminus S_0$.
14. Let $S_0 = \{s_1, \dots, s_{|S_0|}\}$ and for each $s \in S_1$ let $k(s) \in \{0, 1\}$ and $y(s) \subseteq \{1, \dots, |S_0|\}$ such that $s = (-1)^{k(s)} \prod_{i \in y(s)} s_i$, where we fix the order in the product to have increasing indices from left to right.
15. We can now fix the actions of all cycles by projecting onto their ± 1 eigenspaces:
16. Let $x \in \{0, 1\}^{|S_0|}$. We now restrict to a fixed symmetry space, i.e., everything that follows from here on, basically has to be done for each x separately.
17. Restrict to the mutual $+1$ eigenspace $\mathcal{H}[x]$ of $\langle (-1)^{x_i} s_i \mid i \in \{1, \dots, |S_0|\} \rangle$
18. For all $i \in \{1, \dots, |S_0|\}$ choose $b(\eta^{-1}(s_i))$ such that

$$i^{|P(\eta^{-1}(s_i))| \bmod 2} \prod_{e=(u,v) \in P(\eta^{-1}(s_i))} (-1)^{b(e)} i \gamma_u \gamma_v = (-1)^{x_i} , \quad (3.1.8)$$

where the product is ordered with increasing labels from left to right. That is, we have

$$\begin{aligned} (-1)^{b(\eta^{-1}(s_i))} &= (-1)^{x_i} i^{|P(\eta^{-1}(s_i))| \bmod 2} \left(\prod_{e \in P(\eta^{-1}(s_i)) \setminus \{\eta^{-1}(s_i)\}} (-1)^{d(e)} \right) \\ &\quad \times \left(\prod_{e=(u,v) \in P(\eta^{-1}(s_i))} i \gamma_u \gamma_v \right) , \end{aligned} \quad (3.1.9)$$

where the signs on the right-hand have already been fixed in previous steps (the Majorana modes cancel to ± 1 since the product goes over a cycle, and the imaginary units also cancel to ± 1 since there are $|P(\eta^{-1}(s_i))| \bmod 2 + |P(\eta^{-1}(s_i))|$ many of them, which is an even number).

19. Analogously, for all $s \in S_1$ choose $b(\eta^{-1}(s))$ such that

$$i^{|P(\eta^{-1}(s))| \bmod 2} \prod_{e=(u,v) \in P(\eta^{-1}(s))} (-1)^{b(e)} i \gamma_u \gamma_v = (-1)^{k(s) + \sum_{i \in y(s)} x_i}, \quad (3.1.10)$$

20. Finally, for the cycle symmetries proportional to the identity, S_I , we also have to choose a ± 1 eigenspace. However, as the projector onto -1 is zero, we only have to consider the $+1$ eigenspace (which is the full space): For all $s \in S_I$, let $x \in \{0, 1\}$ such that $s = (-1)^x \mathbb{1}$ and choose $b(\eta^{-1}(s))$ such that

$$i^{|P(\eta^{-1}(s))| \bmod 2} \prod_{e=(u,v) \in P(\eta^{-1}(s))} (-1)^{b(e)} i \gamma_u \gamma_v = (-1)^x. \quad (3.1.11)$$

21. ϕ is now completely defined, and importantly, its inverse allows us to define a homomorphism that inverts ϕ on the operators appearing in the problem Hamiltonian, i.e., on V :

22. Set $M = \{(-1)^{b((u,v))} i \gamma_u \gamma_v \mid (u,v) \in F\}$ and define $\mathcal{W}_R = \langle i, g \mid g \in M \rangle$ and $\mathcal{P}[x] = \langle i, g \mid_{\mathcal{H}[x]} \in V \rangle$ for the restricted symmetry space. Note that $\{i\} \cup M$ forms an independent generating set of $\mathcal{W}_R$ as there are no cycles in the spanning tree and none of the Majorana generators squares to i . Furthermore, it is clear that $\phi(V) \subset \mathcal{W}_R$ as for $g \in V \setminus f^{-1}(F)$, $\phi(g)$ can be described by the corresponding cycle $\eta(f(g))$. Now define

$$\begin{aligned} \psi : \mathcal{W}_R &\rightarrow \mathcal{P}[x], \quad i^q \prod_{g \in X} g \mapsto i^q \prod_{e \in X} \phi^{-1}(g) \\ \Leftrightarrow i^q \prod_{e=(u,v) \in Y} (-1)^{b((u,v))} i \gamma_u \gamma_v &\mapsto i^q \prod_{e \in Y} f^{-1}(e), \quad q \in \{0, 1, 2, 3\}, X \subseteq M, Y \subseteq F, \end{aligned} \quad (3.1.12)$$

with the same product order on both sides. This mapping is well-defined (as ϕ preserves the (scalar) commutator) and, per definition, it is a group homomorphism (note that for $g \in \mathcal{W}_R$ we have $g^2 = \mathbb{1}$ as well as $\phi^{-1}(g)^2 = \mathbb{1}$). While it is not necessarily injective (cf. Sec. B.1), it is surjective and it inverts ϕ on V : It is clear that ψ inverts ϕ on $f^{-1}(F)$, per definition. Now, for $g \in V \setminus f^{-1}(F)$ (in the restricted symmetry space), let $s \in S$ be the associated cycle symmetry, i.e., $s = \eta(f(g))$. We have $s = c$ for some $c \in \{-1, +1\}$ in the restricted symmetry space, and therefore

$$g = ac i^{|P(f(g))| \bmod 2} \prod_{e \in P(f(g)) \setminus \{f(g)\}} f^{-1}(e);, \quad (3.1.13)$$

for some appropriate $a \in \{-1, +1\}$ defined by the commutation relations. It follows, with $(u, v) = f(g)$, that

$$\psi(\phi(g)) = \psi \left(\phi(g) c \overbrace{i^{|P(f(g))| \bmod 2}}^{=1} \prod_{e \in P(f(g))} \phi(f^{-1}(e)) \right) \quad (3.1.14a)$$

$$\stackrel{(*)}{=} \psi \left(ac i^{|P(f(g))| \bmod 2} \prod_{e \in P(f(g)) \setminus \{f(g)\}} \phi(f^{-1}(e)) \right) \quad (3.1.14b)$$

$$= ac i^{|P(f(g))| \bmod 2} \prod_{e \in P(f(g)) \setminus \{f(g)\}} \psi(\phi(f^{-1}(e))) \quad (3.1.14c)$$

$$= ac i^{|P(f(g))| \bmod 2} \prod_{e \in P(f(g)) \setminus \{f(g)\}} f^{-1}(e) \quad (3.1.14d)$$

$$= g, \quad (3.1.14e)$$

where in $(*)$ we commuted $\phi(g)$ through the product until it hit itself there - we get the same sign a because ϕ preserves the commutation relations.

Also, importantly note that ψ is invariant w.r.t. hermiticity, i.e., if $g \in \mathcal{W}_R$ is (anti-)hermitian, then $\psi(g)$ is (anti-)hermitian: Let $g \in \mathcal{W}_R$ be hermitian. Both, g and $\psi(g)$ are unitary per definition. Therefore we have $\psi(g)^\dagger = \psi(g)^{-1}$, but also $\psi(g) = \psi(g^{-1}) = \psi(g)^{-1}$, where we used that $g^2 = \mathbb{1}$ because g is unitary and hermitian; thus it we have $\psi(g)^\dagger = \psi(g)^{-1} = \psi(g)$. Analogously for anti-hermitian g .

23. We now extend ψ to an algebra homomorphism, but firstly let us define the according algebras: Let $X \subset \mathcal{W}_R$ be a set of representatives of $\mathcal{W}_R/\langle i \rangle$ such that $\phi(V) \subset X$; analogously, let $Y \subset \mathcal{P}[x]$ be a set of hermitian representatives of $\mathcal{P}[x]/\langle i \rangle$ such that $V \subset Y$. The Hilbert-Schmidt inner product of any two different Majorana strings (ignoring scalar factors), including the identity, is zero; the same holds for any two different operators (ignoring scalar factors) in Y (since they are hermitian and unitary and have vanishing trace). Therefore, X and Y are both (vector-space) linearly independent sets. Define now the complex vectorspaces $W = \text{span}(X)$ and $U = \text{span}(Y)$. Both form an algebra with the multiplication inherited from $\mathcal{W}_R$ and $\mathcal{P}[x]$, respectively (the multiplication is closed since we spanned the vectorspaces with a full set of coset representatives and multiples of i are accounted for by the complex base field).

24. Define the algebra homomorphism

$$\Psi : W \rightarrow U, \quad \sum_{g \in X} a_g g \mapsto \sum_{g \in X} a_g \psi(g), \quad (3.1.15)$$

where $a_g \in \mathbb{C}$ for all $g \in X$. Ψ is clearly well-defined and linear by construction, and it is an algebra homomorphism because ψ is a group homomorphism and preserves multiples of $i\mathbb{1}$. Furthermore, Ψ obviously inverts ϕ on V .

3.2. Diagonalisation of the Majorana Hamiltonian

25. Define the Majorana Hamiltonian

$$H' = \sum_{g \in V} w_g \phi(g) = 2i \sum_{(u,v) \in E'} h_{u,v} \gamma_u \gamma_v = i \sum_{u,v=1}^m h_{u,v} \gamma_u \gamma_v, \quad (3.2.1)$$

where we define $h \in \mathbb{R}^{m \times m}$ to be antisymmetric with $h_{u,v} = w_{f^{-1}(u,v)}(-1)^{b(u,v)}/2$ for $(u,v) \in E'$.

26. If we have multiple components in G , and with that in R , we sort the Majorana modes such that h is block-diagonal with blocks corresponding to these components. The following decompositions of h and diagonalisations could (or should) therefore also be performed on each block separately.

27. As h is antisymmetric and real, there exist an $K \in \text{SO}(m)$ and a $\Lambda \in \mathbb{R}^{m \times m}$ that is block-diagonal with antisymmetric 2×2 blocks or zeros on the diagonal, such that $h = K \Lambda K^{-1}$. These matrices can be found via the Schur decomposition (cf. Cor. C.0.3). *Scipy has a method to perform the Schur decomposition. One might have to do additional work to ensure that $K \in \text{SO}(m)$ and not just $K \in \text{O}(m)$ (cf. the proof of Prop. C.0.2).* Let $J \subseteq \{1, \dots, \lfloor m/2 \rfloor\}$ and $\lambda_j \in \mathbb{R} \setminus \{0\}$, $j \in J$, such that

$$\Lambda = \sum_{j \in J} \begin{pmatrix} 0 & \lambda_j \\ -\lambda_j & 0 \end{pmatrix}_{2j-1, 2j}, \quad (3.2.2)$$

where the subscript indicates the embedding into $\mathbb{R}^{m \times m}$. Note that Λ is block diagonal w.r.t. the components of R , specifically, the λ_j blocks, $j \in J$, do not couple different components.

28. As K is real and orthogonal, we can perform a block-diagonalisation of K into simple rotations on the diagonal (this can again be done via the Schur decomposition (cf. Cor. C.0.5), but we do not actually need the matrices directly). Specifically, there exist an $O \in \text{O}(m)$ and a $D \in \mathbb{R}^{m \times m}$ that is block-diagonal with either ± 1 or 2×2 rotation blocks on the diagonal, such that $K = O D O^\dagger$. As $\det(K) = 1$, we can collect all -1 entries into pairs and parametrise them as 2×2 rotation blocks with rotation angle π (cf. Rem. C.0.6). Set $q = O \ln(D) O^\dagger / 4$, where we choose $\ln(D)$ such that the $+1$ entries map to 0 and the 2×2 rotation blocks map to $\begin{pmatrix} 0 & \theta \\ -\theta & 0 \end{pmatrix}$ for some angle $\theta \in \mathbb{R}$. Now define and $Q = i \sum_{u,v=1}^m q_{u,v} \gamma_u \gamma_v$.

29. The matrix q is clearly real and antisymmetric (because we choose the right logarithm!). Therefore Q is a quadratic Gaussian Hamiltonian, that is, we obtain the usual transformation of the Majorana under the corresponding unitarily evolution:

Define $U(t) = e^{itQ}$, $t \in \mathbb{R}$, and $U = U(1) = e^{-\gamma^\top q \gamma}$. Calculating the derivative of $\gamma_u(t) := U(t)\gamma_u U(t)^\dagger$, $u \in \{1, \dots, m\}$, gives

$$d_t \gamma_u(t) = U(t)i[Q, \gamma_u]U(t)^\dagger \quad (3.2.3a)$$

$$= -U(t) \left(\sum_{v=1}^m [q_{u,v} \gamma_u \gamma_v + q_{v,u} \gamma_v \gamma_u, \gamma_u] \right) U(t)^\dagger \quad (3.2.3b)$$

$$= 4 \sum_{v=1}^m q_{u,v} \gamma_v(t) . \quad (3.2.3c)$$

This is solved by $\gamma_u(t) = \sum_{v=1}^m e^{4tq_{u,v}} \gamma_v$, i.e., for $t = 1$ we have $U\gamma_u U^\dagger = (K\gamma)_u$.

30. We now get

$$UH'U^\dagger = i \sum_{u,v=1}^m h_{u,v} (U\gamma_u U^\dagger) (U\gamma_v U^\dagger) \quad (3.2.4)$$

$$= i \sum_{u,v=1}^m h_{u,v} (K\gamma)_u (K\gamma)_v \quad (3.2.5)$$

$$= i\gamma^\top K^\top h K \gamma \quad (3.2.6)$$

$$= i\gamma^\top \Lambda \gamma \quad (3.2.7)$$

$$= 2i \sum_{j \in J} \lambda_j \gamma_{2j-1} \gamma_{2j} . \quad (3.2.8)$$

Note that the operators in $\{i\gamma_{2j-1}\gamma_{2j}\}_{j \in J}$ are mutually commuting.

31. The operator U is non-trivial, however, since it is the evolution of a quadratic Gaussian Hamiltonian, we can decompose it into a product of $\mathcal{O}(m^2)$ terms of the form $e^{2i\theta\gamma_u\gamma_v}$, $u, v \in \{1, \dots, m\}$, $\theta \in \mathbb{R}$; this is the match gate decomposition [3, Thm. 5]²:
32. Decompose K into $K = K_l \cdots K_1$, $l \in \mathcal{O}(m^2)$ (the swapped order is on purpose here and not with the U_i s below (better consistency with the code)), where each $K_i \in \mathbb{R}^{m \times m}$ is an embedded two dimensional rotation matrix, i.e., K_i acts only non-trivially on the indices $a_i, b_i \in \{1, \dots, m\}$, $a_i \neq b_i$, $i \in \{1, \dots, l\}$. See Chap. D for details.
33. Now, for $i \in \{1, \dots, l\}$ let $\theta_i \in \mathbb{R}$ such that $K_i = e^{4i\theta_i \sigma_{y,a_i b_i}}$ with $\sigma_{y,a_i b_i} \in \mathbb{R}^{m \times m}$ being the embedded Pauli Y matrix acting on the indices a_i and b_i . Set $U_i = e^{-2\theta_i \gamma_{a_i} \gamma_{b_i}} = e^{i\theta_i \gamma^\top \sigma_{y,a_i b_i} \gamma}$. Analogously to above with U , we have

$$U_i \gamma_u U_i^\dagger = (K_i \gamma)_u , \quad (3.2.9)$$

for all $u \in \{1, \dots, m\}$ and $i \in \{1, \dots, l\}$. Therefore it holds that

$$U = U_1 \cdots U_l , \quad (3.2.10)$$

²We don't care for the indices u, v to be only nearest neighbours on a "qubit line", as we do not need that.

up to a phase (which does not matter for us) since the Majorana operators form a complete basis (if necessary include additional Majorana modes on which U and U_i , $i \in \{1, \dots, l\}$, act trivially), and the conjugation action is uniquely defined by its image on a basis.

3.2.1. Diagonalisation and Preparation of Eigenstates of the Problem Hamiltonian

34. From the Majorana diagonalisation, we have

$$(U_1 \cdots U_l) * H' = 2i \sum_{j \in J} \lambda_j \gamma_{2j-1} \gamma_{2j} . \quad (3.2.11)$$

This is an equation in the W algebra and applying Ψ on it gives (making heavy use of the fact that Ψ is an algebra homomorphism)

$$\left(e^{2i\theta_1 \psi(i\gamma_{a_1} \gamma_{b_1})} \cdots e^{2i\theta_l \psi(i\gamma_{a_l} \gamma_{b_l})} \right) * H = 2 \sum_{j \in J} |\lambda_j| \psi(i \operatorname{sign}(\lambda_j) \gamma_{2j-1} \gamma_{2j}) . \quad (3.2.12)$$

As usual, note that the arguments of Ψ are indeed in W , as the diagonalisation procedure kept different graph components separated.

35. The operators $\{\operatorname{sign}(\lambda_j) \psi(i\gamma_{2j-1} \gamma_{2j})\}_{j \in J}$ are mutually commuting, hermitian and unitary, however, they might not be independent; moreover they might also not be independent of S_0 . This makes it in general hard to order the eigenvalues of H , especially finding the ground state energy may not be trivial. We shall give more details on that later below, but first we shall finish the diagonalisation.
36. The operators on the left-hand side of Eq. (3.2.12), i.e., $\Psi(U)$, are exponentials of single operator strings. Depending on the choice of $\mathcal{R}$, these might be easy to decompose into elementary operators, e.g., if $\mathcal{R}$ is the Pauli group, there are standard decompositions (or circuits) for that.
37. Regarding the operators on the right-hand side of Eq. (3.2.12), define $B = \{\psi(i \operatorname{sign}(\lambda_j) \gamma_{2j-1} \gamma_{2j})\}_{j \in J}$ and let $B_0 = \{g_1, \dots, g_t\} \subseteq B$, $t \in \mathbb{N}$, be a maximally independent subset such that $C_0 := S_0 \cup B_0$ is also independent and label $S_0 = \{g_{t+1} = s_1, \dots, g_{t+|S_0|} = s_{|S_0|}\}$. These operators are clearly a set of stabilisers. Therefore, the stabilised states are the according eigenstates of the right-hand side. **With some group theory and symplectic linear algebra, one sees that there exists a generalised Clifford transform operator to implement these states (need to add notes for this ...)**; whether this generalised Clifford transform operator can be realised might depend on the choice of $\mathcal{R}$. Let us specifically consider the case where $\mathcal{R}$ is the Pauli group on n qubits (note that it must be $|C_0| \leq n$): Then, we get a Clifford operator C such that $C * Z_i = g_i$ for all $i \in \{1, \dots, |C_0|\}$. There are papers that show how to realise C , e.g., Refs. [4, 5]; *qiskit also has it implemented*.

38. Set $B_1 = B \setminus B_0$ to be the dependent operators. For $g \in B_1$, define $k(g) \in \{0, 1\}$ and $y(g) \in \{0, 1\}^{|C_0|}$ such that $g = (-1)^{k(g)} \prod_{i=1}^{|C_0|} g_i^{y(g)_i}$. Furthermore, define $j(g) \in J$ such that $g = \psi(i \text{ sign}(\lambda_{j(g)}) \gamma_{2j(g)-1} \gamma_{2j(g)})$. Now, define the diagonal operator

$$D = \sum_{j=1}^t |\lambda_j| Z_j + \sum_{g \in T} |\lambda_{j(g)}| (-1)^{k(g)} \prod_{i=1}^{|C_0|} Z_i^{y(g)_i}. \quad (3.2.13)$$

39. Define the cycle projector ($x \in \{0, 1\}^{|S_0|}$ was the chosen symmetry sector)

$$P[x] = \prod_{j=1}^{|S_0|} \frac{\mathbb{1} + (-1)^{x_j} s_j}{2}. \quad (3.2.14)$$

Note that we have $[P[x], H] = [P[x], \Psi(U)] = 0$ as well as $[P[x], g] = 0$ for all $g \in B$, because all of these operators are defined via operators in V . However, we do not have $[P[x], C] = 0$ in general, but clearly it holds $[P[x], C * D] = 0$ as that is a sum of the stabilisers.

40. It holds that (we know also add the x dependency to $\Psi(U)$, C , D for clarity; there are more dropped dependencies on x , e.g., U and Ψ depend on x , and also t might depend on x (although I believe that t is actually constant))

$$P[x]H = P[x] \left(\left(\Psi(U)[x]^\dagger C[x] \right) * D[x] \right). \quad (3.2.15)$$

Therefore, given a $y \in \{0, 1\}^t$, we can realise the eigenstate as $P[x] \Psi(U)[x]^\dagger C[x] |y\rangle$, where $|y\rangle$ is the computational basis state corresponding to y . Note that we are degenerate on the last $n - t$ qubits.

Note that we cannot “invert” Eq. (3.2.15) to get $P[x]H$ back from $\Psi(U)[x]$, $C[x]$, $D[x]$ and $P[x]$ (except if we do not have any cycle symmetries), as $P[x]$ does not commute with $C[x]$. In full it holds

$$H = \sum_{x \in \{0, 1\}^{|S_0|}} P[x] \left(\Psi(U)[x]^\dagger C[x] * D[x] \right). \quad (3.2.16)$$

41. *The following may not be very rigorous.* Let us now deal with the problem of ordering the eigenvalues in some interval $[\lambda_0, \lambda_0 + \tau]$, $\tau \in \mathbb{R}$ (or alternatively the z smallest eigenvalues) (ignoring the complexity factor corresponding to the number of those energies), where λ_0 is the ground state energy. Firstly, note that the cycle symmetries already introduce $2^{|S_0|}$ non-trivial choices. Furthermore, some of the operators in B may depend on those symmetries. In the following we ignore this dependencies, however, depending on how the (approximate) solution is implemented in code, it is fairly straight forward to include this dependency. We consider two cases:

- (a) The operators are independent. Set $t = |J|$ and for $y \in \{0, 1\}^t$ define the eigenenergies

$$E(y) = 2 \sum_{j=1}^t |\lambda_j| (-1)^{1-y_j} = 2 \left(- \sum_{j=1}^t |\lambda_j| + 2 \sum_{j=1}^t |\lambda_j| y_j \right). \quad (3.2.17)$$

The value of $(-1)^{1-y_j} = -1 + 2y_j$ defines whether we pick the -1 or $+1$ eigenvalue of the operator $\text{sign}(\lambda_j)\psi(i\gamma_{2j-1}\gamma_{2j})$. We want to find, and order, all $y \in \{0,1\}^t$ such that the second summand, $E'(y) := \sum_{j=1}^t |\lambda_j| y_j$ is at most $\tau/2$. This is a standard computational problem (smallest subset sum problem) and can be done efficiently (ignoring $\mathcal{O}(z)$) as follows: *Sort the λ_j s with increasing order and set $A = \{(y = 0, E'(y) = 0)\}$. Now, for $j = 1, \dots, t$, do: set $A' = \{(y' = y + e_j, E'(y')) \mid y \in A\}$, where e_j is the standard Euclidean basis vector; then do a sorted merge of A and A' keeping only those entries with $E'(y) \leq \tau/2$ (or alternatively stop the merge after z entries have been collected); update A to be that merged list; continue the for loop.*

- (b) The operators are not independent. Let $D \subseteq J$ such that $\prod_{j \in D} \psi(i\gamma_{2j-1}\gamma_{2j}) = \pm 1$ ($\pm i1$ is not possible as it is a product of hermitian commuting operators) and set $X = \{2j-1, 2j \mid j \in D\}$. Now consider a component C of R that covers m_C Majorana modes collected in the set M_C . As in Sec. B.2, one sees that either $X \cap M_C = \emptyset$ or $M_C \subseteq X$, and in the latter case it must be m_C even. Specifically, this implies - by considering the case where X is covered by a single component - that any set $\{\psi(i\gamma_{2j-1}\gamma_{2j}) \mid j \in D'\}$ where $D' \subset \{1, \dots, \lfloor m/2 \rfloor\}$ is restricted to a single component C' of G with $|D'| < \lceil |C'|/2 \rceil$ is an independent set. That is, if we do not cover all Majorana modes in the component, the according operators under ψ are independent. We consider two cases now; G being connected or not:

- (1) Let G be connected. In that case we have $D = J = \{1, \dots, \lfloor m/2 \rfloor\}$, and m is even. Let $g_i = \psi(i\gamma_{2i-1}\gamma_{2i})$, $i \in J$, and, w.l.o.g., let $|\lambda_1| \geq \dots \geq |\lambda_{|J|}| > 0$. As we observed that any real subset of $\{g_i \mid i \in J\}$ is independent, we have a closed form for the eigenvalues of H : Set $t = |J| - 1$, $g = \text{sign}(\lambda_{t+1})\psi(i\gamma_{2(t+1)-1}\gamma_{2(t+1)})$, $a = (-1)^{k(g)} \prod_{j=1}^t (-1)^{y(g)_i}$ (cf. Eq. (3.2.13)) and let $z \in \{0,1\}^t$, then

$$E(z) = 2 \sum_{j=1}^t |\lambda_j| (-1)^{1-z_j} + 2|\lambda_{t+1}| a \prod_{j=1}^t (-1)^{1-z_j} \quad (3.2.18)$$

covers all the eigenvalues of H . Clearly, the last term only depends on the parity of y . *Therefore, we can solve the problem analogously as in the case of independent operators by having two lists A_{even} and A_{odd} for even and odd parity of y , respectively, and getting a new $A'_{\text{even (odd)}}$ from $A_{\text{odd (even)}}$ and sorted merging it with $A_{\text{even (odd)}}$. After the end of the for loop, we just sorted merge the two lists.*

- (2) Let G be not connected. The problem is that the independent generators from one component might influence not just the single remaining eigenvalue in that component but also the single remaining eigenvalue in other components. Therefore the approach above fails, from a complexity point of view, because we would have to consider multiple parities and the scaling would be exponential in the number of those parities (*exponential number of the A lists*).

We see that we may run into problems if $|S_0|$ plus the number of graph components is too big. Depending on that number one may choose an approximated search for the ground state energy. For example, just pick the y s according to a Metropolis-Hastings sampling from the distribution $\pi(y) \propto e^{-\beta E(y)}$, where the coldness function $\beta \in \mathbb{R}_{>0}$ is a parameter that controls how much we focus on low energies.

APPENDICES

A. Minor Things

Proposition A.0.1 *Let $g_1, \dots, g_m \in \mathcal{R}$ be hermitian, $m \in \mathbb{N}$, such that the according frustration graph is a cycle, i.e., g_i and g_j anticommute if, and only if, $|i - j| = 1 \bmod m$. Then, $g_1 \cdots g_m$ is hermitian if m is even and antihermitian if m is odd.*

Proof. We have

$$(g_1 \cdots g_m)^\dagger = g_m \cdots g_1 \quad (\text{A.0.1a})$$

$$= (-1)^2 g_1 g_m \cdots g_2 \quad (\text{A.0.1b})$$

$$= (-1)^{2+1} g_1 g_2 g_m \cdots g_3 \quad (\text{A.0.1c})$$

$$= (-1)^{2+2} g_1 g_2 g_3 g_m \cdots g_4 \quad (\text{A.0.1d})$$

$$\vdots \quad (\text{A.0.1e})$$

$$= (-1)^{2+m-2} g_1 \cdots g_m \quad (\text{A.0.1f})$$

$$= (-1)^m g_1 \cdots g_m . \quad (\text{A.0.1g})$$

□

Proposition A.0.2 *Let $g, h_1, \dots, h_m \in \mathcal{R}$, $m \in \mathbb{N}$. Then it holds*

$$\llbracket g, h_1 \cdots h_m \rrbracket = \llbracket g, h_1 \rrbracket \cdots \llbracket g, h_m \rrbracket . \quad (\text{A.0.2})$$

Proof. This follows inductively. The case $m = 1$ is clear. Now let the statement be true for $m - 1$ and set $h = h_2 \cdots h_m$. Then we get

$$\llbracket g, h_1 \cdots h_m \rrbracket = \llbracket g, h_1 h \rrbracket \quad (\text{A.0.3})$$

$$= g h_1 h g^{-1} h^{-1} h_1^{-1} \quad (\text{A.0.4})$$

$$= g h_1 g^{-1} h_1^{-1} h_1 g h g^{-1} h^{-1} h_1^{-1} \quad (\text{A.0.5})$$

$$= \llbracket g, h_1 \rrbracket h_1 \llbracket g, h \rrbracket h_1^{-1} \quad (\text{A.0.6})$$

$$= \llbracket g, h_1 \rrbracket \llbracket g, h \rrbracket \quad (\text{A.0.7})$$

$$= \llbracket g, h_1 \rrbracket \cdots \llbracket g, h_m \rrbracket . \quad (\text{A.0.8})$$

□

B. Symmetries Under the Ψ and ψ Mappings

B.1. Why ψ is not Injective in General

Consider the following frustration graph in the Pauli basis

$$X_1 X_2 X_3 X_4 Z_5 \text{ --- } Y_5 \text{ --- } X_1 X_2 Z_5 \text{ --- } Y_2 Y_3 \text{ --- } X_3 X_4$$

with the root graph

$$\gamma_1 \text{ --- } \gamma_2 \text{ --- } \gamma_3 \text{ --- } \gamma_4 \text{ --- } \gamma_5 \text{ --- } \gamma_6.$$

There are clearly no cycles symmetries. ψ maps $i\gamma_1\gamma_2$ to $X_1X_2X_3X_4Z_5$ but we also have $\psi(-\gamma_3\gamma_4\gamma_5\gamma_6) = \psi(i\gamma_3\gamma_4)\psi(i\gamma_5\gamma_6) = X_1X_2X_3X_4Z_5$. In general, the reason why we cannot have an homomorphism from $\mathcal{P}[x]$ to $\mathcal{W}_R$, which preserves ϕ , is that even if we fix cycle symmetries, there can be vertices in G that are not related by a cycle but still are not independent to each other. This does not happen with edges in spanning tree of the root graph, i.e., the Majorana hopping terms, because they are all independent per construction.

However, dependent Hamiltonian terms must correspond to a T-join (if not they are not a cycle) as we shall see in the next section.

B.2. Elements in $\psi^{-1}(\mathbb{1})$ must be an Effective T-join (or a Cycle)

Let $g \in \mathcal{W}_R$ with $\psi(g) = \mathbb{1}$, up to a phase. The set $B = \{(-1)^{d((u,v))}i\gamma_u\gamma_v \mid (u,v) \in E'\}$ is clearly a generating set of $\mathcal{W}_R$ (but not necessarily independent). Therefore, w.r.t. to that generating set, if $g = \mathbb{1}$, up to a phase, g corresponds to a cycle (or multiple ones) in R (and/or single edges an even number of times). Now assume that $g = \prod_{i \in X} \gamma_i$, up to a phase, for some non-empty $X \subseteq \{1, \dots, m\}$. Fix a component C of R which covers, w.l.o.g., the first $m_C \in \mathbb{N}$, $m_C \leq m$, Majorana modes. Clearly, for any $u, v \in \{1, \dots, m_C\}$, $u \neq v$, we have $a := \gamma_u\gamma_v \in \mathcal{W}_R$. As ψ is a homomorphism, we have $\psi(\llbracket a, g \rrbracket) = \mathbb{1}$, and because we also know that $\llbracket a, g \rrbracket = \pm \mathbb{1}$ and $\psi(-\mathbb{1}) = -\mathbb{1}$, we have $\llbracket a, g \rrbracket = \mathbb{1}$. Therefore, as u, v were arbitrary it must be either that $X \cap \{1, \dots, m_C\} = \emptyset$ or $\{1, \dots, m_C\} \subseteq X$ (basic Majorana string commutation rules). In the latter case, g therefore, with respect to the generating set B , corresponds to a T-join in the component C and m_C must be even (and the former case could be a cycle again).

B.3. How to Make ψ Injective

This is not important for us, but with some thought one sees (considering the previous section) that if we restrict in the Majorana picture to symmetry subspaces of the T-joins (one per component), ψ becomes injective and with that an isomorphism.

C. On the Schur Decomposition

Definition C.0.1 (Quasi-triangular matrix) *A matrix $T \in \mathbb{R}^{n \times n}$ is called quasi-triangular if it is block-upper-triangular with blocks of size at most 2×2 on the diagonal.*

Proposition C.0.2 (Real Schur decomposition) *Let $A \in \mathbb{R}^{n \times n}$, $n \in \mathbb{N}$. Then, there exist an $M \in \text{SO}(n)$ and a $T \in \mathbb{R}^{n \times n}$ that is quasi-triangular, such that $A = MTM^{-1}$.*

Proof. This is a standard result in linear algebra. However, as some implementations only yield $M \in \text{O}(n)$, we give a short construction to ensure that $M \in \text{SO}(n)$: Let $A = M'T'M'^{-1}$ be a Schur decomposition of A , i.e., $M' \in \text{O}(n)$ and $T' \in \mathbb{R}^{n \times n}$ is quasi-triangular. If $\det(M') = 1$, we are done. Otherwise, define $D = \text{diag}(-1, 1, \dots, 1) = \mathbb{1} - 2(\delta_{i,1}\delta_{j,1})_{i,j=1}^n \in \mathbb{R}^{n \times n}$ and set $M = M'D$ and $T = DT'D$. Clearly, we have $A = MTM^{-1}$, $M \in \text{SO}(n)$ and T is quasi-triangular as well. $\square$

Corollary C.0.3 *Let $A \in \mathbb{R}^{n \times n}$ be antisymmetric, $n \in \mathbb{N}$. Then, the Schur decomposition yields an $M \in \text{SO}(n)$ and a $B \in \mathbb{R}^{n \times n}$ that is block-diagonal with antisymmetric 2×2 blocks or zeros on the diagonal, such that $A = MBM^{-1}$.*

Proof. Let $A = MBM^{-1}$ be a Schur decomposition of A , i.e., $M \in \text{SO}(n)$ and $B \in \mathbb{R}^{n \times n}$ is quasi-triangular. We have

$$B^\top = (M^{-1}AM)^\top = -M^{-1}AM = -B, \quad (\text{C.0.1})$$

so B is antisymmetric and the claim follows. $\square$

Remark C.0.4 *If A is block-diagonal (with blocks of arbitrary size), the matrices M and T (B) in the Schur decomposition are accordingly block-diagonal, i.e., the decomposition holds for each block separately.*

It is known that real orthogonal matrices can be block-diagonalised into 2×2 rotation matrices and reflections. The Schur decomposition also achieves that:

Corollary C.0.5 *Let $A \in \text{O}(n)$, $n \in \mathbb{N}$. Then, the Schur decomposition yields an $M \in \text{SO}(n)$ and an $R \in \mathbb{R}^{n \times n}$ that is block-diagonal with 2×2 rotation matrices or ± 1 entries on the diagonal, such that $A = MRM^{-1}$.*

Proof. Let $A = MRM^{-1}$ be a Schur decomposition of A , i.e., $M \in \text{SO}(n)$ and $R \in \mathbb{R}^{n \times n}$ is quasi-triangular; then

$$R^\top = (M^{-1}AM)^\top = M^{-1}A^\top M = M^{-1}A^{-1}M = (M^{-1}AM)^{-1} = R^{-1}, \quad (\text{C.0.2})$$

that is, R is also orthogonal. We now proof the statement by induction over n : For $n = 1$ or $n = 2$ the statement is clear per definition. Now, let $n > 2$ and the statement be true for $n - 1$ and $n - 2$. There are two cases to consider:

Firstly, consider the case where the first column of R has only one non-zero entry, i.e., $R_{1,1} \neq 0$ and $R_{i,1} = 0$ for all $i > 1$. Because the column must be normalised, we have $R_{1,1} = \pm 1$. Furthermore, because the column must be orthogonal to all other columns, we have $R_{1,j} = 0$ for all $j > 1$. The remaining $(n - 1) \times (n - 1)$ lower-right block of R

is again orthogonal and quasi-triangular, so the statement follows from the induction hypothesis.

Secondly, consider the case where the first column of R has two non-zero entries, i.e., $R_{1,1}, R_{2,1} \neq 0$ and $R_{i,1} = 0$ for all $i > 2$. The normalisation implies that there is a $\phi \in [0, 2\pi)$, with $\phi \notin \{\pi/2, 3\pi/2\}$, such that $R_{1,1} = \cos(\phi)$ and $R_{2,1} = \sin(\phi)$. The orthogonality to the second column (and its normalisation) implies that $R_{1,2} = -\sin(\phi)$ and $R_{2,2} = \cos(\phi)$. All the other columns must be orthogonal to the first two columns, so for all $i > 2$ we have

$$\begin{pmatrix} \cos(\phi) & \sin(\phi) \\ -\sin(\phi) & \cos(\phi) \end{pmatrix} \begin{pmatrix} R_{1,i} \\ R_{2,i} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad (\text{C.0.3})$$

but then $R_{1,i} = R_{2,i} = 0$ for all $i > 2$ because this matrix has full rank. The statement now follows from the induction hypothesis applied to the remaining $(n-2) \times (n-2)$ lower-right block of R . $\square$

Remark C.0.6 (a) Taking the logarithm of an orthogonal matrix A via this decomposition yields $\ln(A) = M \ln(R) M^{-1}$ where canonical $\ln(R)$ is block-diagonal with 2×2 real antisymmetric blocks, zeros or $i\pi$ entries on the diagonal. Thus the logarithm is not guaranteed to be real because of potential reflections in R ; the simplest counter example is $A = -\text{id}$.

(b) However, if $A \in \text{SO}(n)$, we can choose the logarithm of R such that there are no $i\pi$ entries on the diagonal: Simply pair up all -1 entries into 2×2 blocks and parametrise them as rotation matrices with angle π . The logarithm such a block gives $\pi \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

D. On the Generalised Euler Angles Decomposition

Reference. [6] gives a generalisation of the Euler decomposition for orthogonal matrices. Specifically, section 4 gives an algorithm on how to obtain generalised Euler angles. We want to decompose this decomposition further into two-dimensional rotations. While Ref. [3] more or less indirectly states that the paper also provides a decomposition into two-dimensional rotations, I cannot find it explicitly in the paper and it also seems not that trivial to me. The following is written as complementary information to Ref. [6], using the same notation, where we refer to equations in Ref. [6] with labels of the form “(EqXX)”:

Let $N \in \mathbb{N}$, $T \in \text{SO}(N)$, $N \in \mathbb{N}$. Then we have $T = B^N \cdots B^2$ (Eq22), with $B^\nu \in \text{SO}(N)$ defined by an $A^\nu \in \text{SO}(\nu)$ on the upper-left $\nu \times \nu$ block and the identity on the remaining diagonal (zeros everywhere else) (Eq16-19,22) for $\nu = 2, \dots, N$. We now decompose each A^ν into two-dimensional rotations:

Proposition D.0.1 Let $\nu \in \{2, \dots, N\}$ and define $R^\mu \in \text{SO}(\nu)$, $\mu \in \{1, \dots, \nu - 1\}$, via

$$R_{\mu,\mu}^\mu = R_{\nu,\nu}^\mu = c_\mu^\nu \quad (\text{D.0.1})$$

$$R_{\mu,\nu}^\mu = -R_{\nu,\mu}^\mu = s_\mu^\nu \quad (\text{D.0.2})$$

$$R_{i,i}^\mu = 1 \quad \text{for } i \in \{1, \dots, \nu\} \setminus \{\mu, \nu\} \quad (\text{D.0.3})$$

and the other entries to be zero, where $c^\nu, s^\nu \in \mathbb{R}^{\nu-1}$ as in (Eq23). Then, it holds

$$A^\nu = R^{\nu-1} R^{\nu-2} \dots R^1. \quad (\text{D.0.4})$$

Proof. For $\nu \in \{2, \dots, N\}$ and $c, s \in \mathbb{R}^{\nu-1}$ define $A(\nu, c, s) \in \text{SO}(\nu)$ via

$$\begin{aligned} A(\nu, c, s)_{i,i} &= c_i & \text{for } i \in \{1, \dots, \nu - 1\} \\ A(\nu, c, s)_{i,\nu} &= c_1 c_2 \dots c_{i-1} s_i & \text{for } i \in \{1, \dots, \nu\} \\ A(\nu, c, s)_{i,k} &= -c_{k+1} c_{k+2} \dots c_{i-1} s_k s_i & \text{for } k \in \{1, \dots, \nu - 1\}, i \in \{k + 1, \dots, \nu\} \\ A(\nu, c, s)_{i,k} &= 0 & \text{for } k \in \{1, \dots, \nu - 1\}, i \in \{1, \dots, k - 1\} \end{aligned} \quad (\text{D.0.5})$$

Clearly, it holds $A^\nu = A(\nu, c^\nu, s^\nu)$, for all $\nu \in \{2, \dots, N\}$. Analogously, we define $R^\mu(\nu, c, s) \in \text{SO}(\nu)$ as above in Eqs. (D.0.1) to (D.0.3) in terms of $c, s \in \mathbb{R}^{\nu-1}$, i.e., it holds $R^\mu = R^\mu(\nu, c^\nu, s^\nu)$ for $\mu \in \{1, \dots, \nu - 1\}$. We now proof the statement by induction over ν but with respect to the parametrised matrices $A(\nu, c, s)$ and $R(\nu, c, s)$, i.e., we show that $A(\nu, c, s) = R^1(\nu, c, s) R^2(\nu, c, s) \dots R^{\nu-1}(\nu, c, s)$ for $\nu \in \{2, \dots, N\}$:

The base case $\nu = 2$ is clear. Now, let $\nu > 2$ and the statement be true for $\nu - 1$. Define $c' = (c_2, \dots, c_{\nu-1}) \in \mathbb{R}^{\nu-2}$ and $s' = (s_2, \dots, s_{\nu-1}) \in \mathbb{R}^{\nu-2}$. Then we have

$$A(\nu - 1, c', s') = R^{\nu-2}(\nu - 1, c', s') \dots R^1(\nu - 1, c', s'), \quad (\text{D.0.6})$$

and therefore

$$\begin{pmatrix} 1 & 0 \\ 0 & A(\nu - 1, c', s') \end{pmatrix} = R^{\nu-1}(\nu, c, s) \dots R^2(\nu, c, s). \quad (\text{D.0.7})$$

It follows

$$R^{\nu-1}(\nu, c, s) \dots R^1(\nu, c, s) = \begin{pmatrix} 1 & 0 \\ 0 & A(\nu - 1, c', s') \end{pmatrix} R^1(\nu, c, s) \quad (\text{D.0.8a})$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & A(\nu - 1, c', s') \end{pmatrix} \begin{pmatrix} c_1 & 0 & s_1 \\ 0 & \mathbb{1}_{\nu-2} & 0 \\ -s_1 & 0 & c_1 \end{pmatrix} \quad (\text{D.0.8b})$$

$$= A(\nu, c, s). \quad (\text{D.0.8c})$$

□

Bibliography

1. Chapman, A. & Flammia, S. T., Characterization of solvable spin models via graph invariants. [Quantum 4, 278 \(2020\)](#), [arXiv](#).

2. Beineke, L. W., Characterizations of derived graphs. [Journal of Combinatorial Theory](#) **9**, 129–135 (1970).
3. Jozsa, R. & Miyake, A., Matchgates and classical simulation of quantum circuits. [Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences](#) **464**, 3089–3106 (2008).
4. Aaronson, S. & Gottesman, D., Improved simulation of stabilizer circuits. [Physical Review A](#) **70**, (2004).
5. Bravyi, S., Shaydulin, R., Hu, S. & Maslov, D., Clifford Circuit Optimization with Templates and Symbolic Pauli Gates. [Quantum](#) **5**, 580 (2021).
6. Hoffman, D. K., Raffinetti, R. C. & Ruedenberg, K., Generalization of Euler Angles to N-Dimensional Orthogonal Matrices. [Journal of Mathematical Physics](#) **13**, 528–533 (1972).